THE SPACE JOURNAL

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APOD: 2026 October 31 - Night on a Spooky Planet

What spooky planet is this? Planet Earth on a dark and stormy night, of course. The ghostly scene was photographed at Hverir , a geothermally active area found along northeastern Iceland's volcanic landscape.

THE EDITION

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A considered selection of source-linked reports from 01 November 2026. Stories retain their original reporting context and source details.

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A NOTE FROM THE EDITORS

This digital edition is composed as a traditional book: one page at a time, with a quiet reading rhythm, printed page numbers and source notes. The production technology stays behind the pages.

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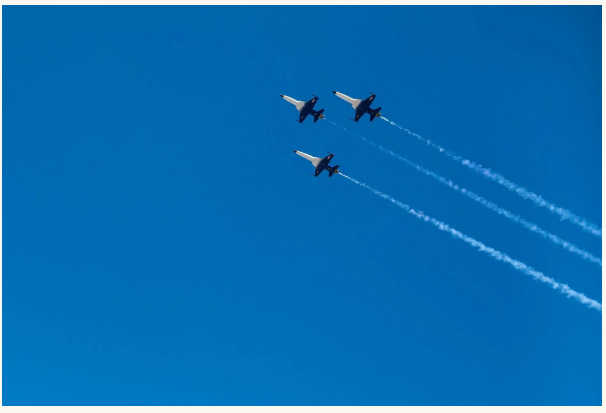


spooky / Stéphane Vetter (Nuits sacrées)

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What spooky planet is this? Planet Earth on a dark and stormy night, of course. The ghostly scene was photographed at Hverir , a geothermally active area found along northeastern Iceland's volcanic landscape . Triggered by solar activity , geomagnetic storms have unleashed energetic particles trapped in Earth's magnetic field. Penetrating the planet's upper atmosphere at high latitudes, the particles create auroral displays in the starry night sky. Ghostly towers of steam and gas are venting from local fumaroles to dance against the eerie, greenish northern lights. For now, auroral activity on planet Earth remains in a high state as the Sun is only in early phases of decline from the peak of its solar activity cycle . And soon, other ghostly apparitions may dance in your neighborhood. Happy Halloween !

NASA Releases MAX POWER Key Events for Media, Participants



Looking up, three white and black fighter jets fly in formation across a bright, blue sky. / NASA

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NASA's Freedom 250 F-5 team, including NASA Administrator Jared Isaacman, performs a flyover during the national anthem at an NFL game between the Cleveland Browns and the Carolina Panthers at Huntington Bank Field, Sept. 27, 2026, in Cleveland. Credit: NASA/Jordan Cochran America's newest aerospace expo and air show - MAX POWER - is fast approaching, and NASA has announced key events beginning Friday, Nov. 6, and running through Sunday, Nov. 8, hosted on and near the agency's Kennedy Space Center in Florida. Through multiple park experiences at MAX POWER , NASA and a variety of external collaborators will showcase the newest aircraft, autonomous vehicles, technologies, and more defining the future of air and space. NASA will display its quiet supersonic X-59 plane, and agency partners will showcase Moon rovers, spacecraft, and more. The expo also is a culmination of the agency's recent Inspiration Tour , which included numerous flyovers to highlight NASA's fleet of aircraft. A current schedule of events is available on the agency's website . MAX POWER includes opportunities to meet leadership, astronauts, engineers, scientists, and other subject matter experts during the family-friendly public events with gates opening at 9 a.m. EST on Nov. 7 and Nov. 8. NASA also will share multiple news updates during the weekend, including hosting a news conference at 5 p.m. on Nov. 7, to provide its latest Moon Base update. Located near the lunar South Pole, the Moon Base is humanity's first outpost for long-term lunar exploration. Live coverage of the news conference will stream on a variety of platforms online. Learn where to watch: https://www.nasa.gov/live

NASA Releases MAX POWER Key Events for Media, Participants

Join NASA for Key Events Media credentialing for this event is closed. Media approved for this event must request interviews in advance. Space in the news conference is limited, and first come, first served. Media also may participate in the news conference virtually. To request call details or an interview, please contact Lauren Low in the NASA Headquarters newsroom at: lauren.e.low@nasa.gov . A short list of key NASA activities during MAX POWER are listed below. All times are listed in Eastern Standard Time, and are subject to change. Friday, Nov. 6: 2 p.m., to 4 p.m.: Media one-on-one opportunities to preview weekend MAX POWER activities Location: NASA Press Site Saturday, Nov. 7: 10 a.m.: Nuclear NASA media gaggle Location: UP.Summit Main Stage Cynthia Simmons, director, Space Reactors Division 11:30 a.m.: Opening air show remarks featuring NASA Deputy Administrator Matt Anderson Location: Air Dot Show Main Stage 11:58 a.m.: F-5 flyover in the Air Dot Show featuring NASA Administrator Jared Isaacman 12:45 p.m.: Meet the Astronauts: NASA's Artemis III Crew Location: Air Dot Show Main Stage Participants include: NASA astronaut Randy Bresnik (mission commander) ESA (European Space Agency) astronaut Luca Parmitano (pilot) NASA astronaut Frank Rubio (mission specialist) NASA astronaut Bob Hines, backup crew member NASA will host a media gaggle with crew immediately following this event. 1:31 p.m.: MiG-29 flyover featuring Isaacman 2 p.m.: Learn about NASA's Roman Space Telescope Location: Kennedy Space Center Visitor Complex (Journey to Mars Theater) Participant: Remarks by Julie McEnery, senior project scientist, Nancy Grace Roman Space Telescope, NASA Goddard 5 p.m. Moon Base News Conference Location: Press Site AuditoriumParticipants: NASA Administrator Jared IsaacmanCarlos García-Galán, manager, Moon Base ProgramNicky Fox, associate administrator for Science, NASA HeadquartersDayna Ise, acting director, Utility and Infrastructure Office, Moon Base Program NASA subject matter experts will be available for one-on-one interviews following this announcement. Sunday, Nov. 8 9 a.m. NASA's X-59 media gaggle Location: UP.Summit Exhibit Area Participants: Laurie Grindle, director, NASA Aeronautics Division Peter Coen, NASA Quesst mission manager, NASA Langley Jim "Clue" Less, test pilot, X-59, NASA Armstrong 10 a.m.: Student challenge announcement Location: Kennedy Space Center Visitor Complex, Rocket Garden Participants: NASA Administrator Jared Isaacman

NASA Releases MAX POWER Key Events for Media, Participants

Carlos García-Galán, manager, Moon Base Program NASA experts may be available for interviews at this location. Media interested in attending must contact Lauren Low by 5 p.m. on Thursday, Nov. 5, at: lauren.e.low@nasa.gov . 11:58 a.m.: F-5 flyover in the Air Dot Show featuring Isaacman 12:45 p.m.: United States Space Academy Location: UP.Summit Main Stage NASA Deputy Administrator Matt Anderson 12:45 p.m.: Meet the Astronauts: NASA's SpaceX Crew-12 Location: Air Dot Show Main Stage Participants: NASA astronaut Jessica Meir NASA astronaut Jack Hathaway 1:31 p.m.: MiG-29 flyover featuring Isaacman For more information about MAX POWER, including the latest schedule of events and how to attend, visit the agency's website: https://www.nasa.gov/maxpower -end- George Alderman / Cheryl Warner Headquarters, Washington 202-358-1600 george.a.alderman@nasa.gov / cheryl.m.warner@nasa.gov

Space Crops and Advanced Tech Wrap Up Week Following Spacewalk



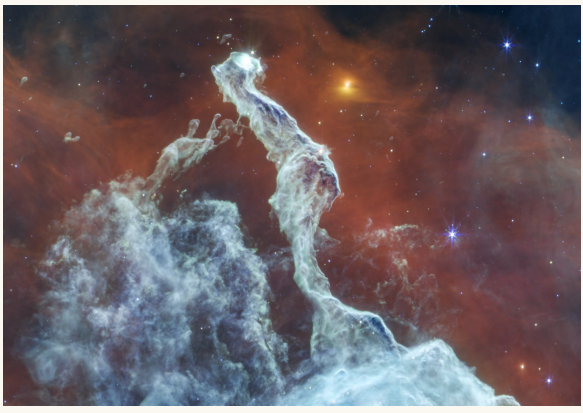
Expedition 75 flight engineers Joshua Kutryk of CSA (Canadian Space Agency) and Jessica Watkins of NASA pose together inside the International Space Station's Quest airlock. Watkin / NASA

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Expedition 75 flight engineers Joshua Kutryk of CSA (Canadian Space Agency) and Jessica Watkins of NASA pose together inside the International Space Station's Quest airlock during a spacesuit fit check. NASA/Anil Menon Space crops research wrapped up the week following the completion of the first of two power upgrade spacewalks at the International Space Station . The Expedition 75 crew members rounded out their research schedule with virtual reality and super intelligence studies. The orbital outpost has a new roll-out solar array that is generating power after flight engineers Anil Menon of NASA and Joshua Kutryk of CSA (Canadian Space Agency) installed it during a spacewalk on Thursday . Menon and Kutryk spent seven hours and 11 minutes in the vacuum of space attaching the roll-out solar array to the port side truss structure then releasing its tension bolts allowing the solar array to unfurl, or roll out, on its own during a ten-minute process. The pair will install one more roll-out solar array on Thursday, Nov. 5, to complete the yearslong process that began in June 2021 to augment the space station's power generation system. Menon and Kutryk will install the eighth and final roll-out solar array on the starboard side of the truss structure increasing the station's power generating capacity from 160 kilowatts to about 215 kilowatts. Menon and Kutryk had a light duty day on Friday with ongoing research planned for their short schedule. Menon worked on a space botany investigation in the Kibo laboratory module exploring how microgravity affects cell division on green algae and cultured tobacco cells. Results may support the

production of food crops on the Moon and Mars. Kutryk pedaled on the European Enhanced Exploration Exercise Device's (E4D) exercise cycle in the Columbus laboratory module . The compact all-in-one E4D is being tested for its ability to keep crews fit during a long-term spaceflight. NASA flight engineer Luke Delaney also studied space-based crops focusing on how to prevent plant diseases in microgravity. Delaney inoculated thale cress plant samples after they were exposed to a fungal pathogen then stowed the samples inside an artificial gravity-generating incubator . Results may improve crop disease resistance both on Earth and in space. NASA flight engineer Jessica Watkins spent her shift servicing a variety of research hardware. She first removed and photographed components inside a device that powers experiments launched aboard spacecraft. Next, Watkins swapped moisture absorbers inside a set of portable science freezers that can be launched to the station and returned back to Earth housing frozen research samples for analysis. Station commander Pyotr Dubrov of Roscosmos worked on a physics experiment swapping hardware inside a temperature-controlled chamber to explore manufacturing high-purity semiconductor crystals in microgravity. Flight engineer Sergey Teteryatnikov wore a virtual reality headset and responded to computerized visual stimuli to testing how his sense of balance and orientation is adapting to microgravity. Flight engineer Anna Kikina studied how super intelligence tools may increase operational and communications efficiency aboard a spacecraft. Learn more about station activities by following the space station blog , @space_station on X, as well as the ISS Facebook and ISS Instagram accounts. Get the latest from NASA delivered every week. Subscribe here .

NASA's James Webb Space Telescope Captures Eerie Cat's Head

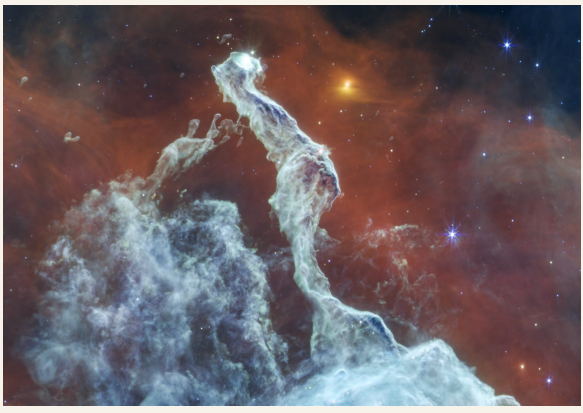


Mid-infrared image of the Cat's Head Nebula by the James Webb Space Telescope. A bluish-gray filament of dust is in the center, stretching from the bottom to almost the very top. I / NASA

30 OCT 2026

NASA's James Webb Space Telescope's mid-infrared view reveals intricate structures within the Cat's Head Nebula, traced by the emission from warm dust and carbon-based molecules excited by starlight.

NASA's James Webb Space Telescope Captures Eerie Cat's Head



Mid-infrared image of the Cat's Head Nebula by the James Webb Space Telescope. A bluish-gray filament of dust is in the center, stretching from the bottom to almost the very top. I / NASA

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NASA, ESA, CSA, STScI; Image Processing: Joseph DePasquale (STScI) Intricate structures within the Cat's Head Nebula are illuminated in this mid-infrared view from NASA's James Webb Space Telescope released on Oct. 29, 2026. The distinctive "cat's head" shape of the nebula was likely sculpted over time by the intense ultraviolet radiation and winds from massive stars in the Carina Nebula. As this stellar feedback erodes the surrounding gas and dust, denser material within the feline-shaped pillar resists erosion, at least for now. Image credit: NASA, ESA, CSA, STScI; Image Processing: Joseph DePasquale (STScI)

NASA Explores Potential Commercial Servicing for Hubble Telescope

30 OCT 2026

The Hubble Space Telescope returns to orbit as an improved telescope after its second servicing mission in February 1997. NASA NASA issued a Request for Information Oct. 29 to learn more about industry capabilities for in-space servicing to support the agency's Hubble Space Telescope. "Hubble still has important science ahead of it, and we should be looking at every reasonable opportunity to keep it delivering for as long as possible," said NASA Administrator Jared Isaacman. "If there are innovative and cost-effective ways to extend the life of an asset already on orbit, we owe it to the American taxpayer and the scientific community to explore them. Extending Hubble's life would mean more discoveries, more science, and more opportunities to expand our understanding of the universe." The Request for Information seeks commercial interest in: potentially conducting a mission to boost Hubble to a higher altitude to extend its science lifetime; servicing Hubble to enhance its capabilities; or performing a safe, controlled re-entry into Earth's atmosphere or orbital disposal at the end of Hubble's mission. "Hubble is a national icon that has fundamentally changed how humanity views our universe for more than 35 years," said Nicky Fox, associate administrator, Science Mission Directorate, NASA Headquarters in Washington. "Commercial industry has rapidly evolved in recent years to offer new capabilities for in-space satellite capture and servicing. Given the time we have until Hubble's anticipated re-entry, we are eager to explore the breadth of possibilities to boost Hubble to a higher orbit while we develop the flagship mission to follow." The Request for Information is available online and will remain open for responses until 3 p.m. EST, Monday, Nov. 30. Like all spacecraft in low Earth orbit, Hubble's altitude is slowly decreasing over time because of atmospheric drag. Hubble does not have a propulsion system to maintain its own orbit, and analysis predicts Hubble is likely to re-enter the atmosphere in the 2033 to 2034 timeframe. As such, NASA anticipates that a mission would need to take place by 2032. The Hubble Space Telescope has been operating for more than three decades and continues to make ground-breaking discoveries that shape our fundamental understanding of the universe. Hubble is a project of international cooperation between NASA and ESA (European Space Agency). NASA's Goddard

NASA Explores Potential Commercial Servicing for Hubble Telescope

Space Flight Center in Greenbelt, Maryland, manages the telescope and mission operations. Lockheed Martin Space, based in Denver, also supports mission operations at Goddard. The Space Telescope Science Institute in Baltimore, which is operated by the Association of Universities for Research in Astronomy, conducts Hubble science operations for NASA. To learn more about NASA's Hubble Space Telescope, visit: https://www.nasa.gov/hubble Facebook logo @NASAHubble @NASAHubble Instagram logo @NASAHubble Media Contacts: Alise Fisher Headquarters, Washington 202-358-2546 alise.m.fisher@nasa.gov Claire Andreoli Goddard Space Flight Center, Greenbelt, Md. 202-286-1940 claire.andreoli@nasa.gov Share Details Last Updated Oct 30, 2026 Related Terms Hubble Space Telescope Astrophysics Astrophysics Division Goddard Space Flight Center The Universe Keep Exploring Discover More Topics From NASA Hubble Space Telescope Since its 1990 launch, the Hubble Space Telescope has changed our fundamental understanding of the universe. Nancy Grace Roman Space Telescope The Nancy Grace Roman Space Telescope will settle essential questions in the areas of dark energy, exoplanets, and astrophysics. TESS (Transiting Exoplanet Survey Satellite) NASA's TESS discovers exoplanets, worlds beyond our solar system. In the course of its extended observations of the sky, TESS... James Webb Space Telescope Webb is the premier observatory of the next decade, serving thousands of astronomers worldwide. It studies every phase in the...

November's Night Sky Notes: Light-Day Away

30 OCT 2026

2 min read November's Night Sky Notes: Light-Day Away NASA's Voyager mission took advantage of a once-every-175-year alignment of the outer planets for a grand tour of the solar system. The twin spacecraft revealed details about Jupiter, Saturn, Uranus and Neptune - using each planet's gravity to send them on to the next destination. Voyager set the stage for such ambitious orbiter missions as Galileo to Jupiter and Cassini to Saturn. Today both Voyager spacecraft continue to return valuable science from the far reaches of our solar system. NASA/JPL-Caltech Twin Travelers In 1977, NASA launched twin spacecraft Voyager 1 and 2, taking advantage of a once-every-175-year planetary alignment of the four largest planets in our solar system: Jupiter, Saturn, Uranus, and Neptune. Using six gravity assists between them, this Grand Tour gave us up-close views of the gas and ice giants, including several moons. By the 1980s, the two probes went their separate ways at Saturn. As Voyager 2 continued toward Uranus and Neptune, Voyager 1 grabbed one more portrait of the solar system (excluding Mercury and Mars), including a final image of the pale blue dot we call home. The Golden Record Cover NASA/JPL-Caltech Playlist "Hello from the children of planet Earth." Every good road trip needs some music to match. In this case, each craft carries the Golden Record , a 12-inch phonograph record with spoken greetings in 55 languages, 27 musical tracks, 116 images, and other sounds from Earth. Each record is protected by an outer cover featuring etchings showing our Sun's location relative to 14 pulsars, along with instructions on how to play it. The design used various metals to protect the information encoded from space radiation: the record used copper and gold plating, and the cover used aluminum. The cover was electroplated with an ultra-pure sample of the isotope uranium-238. Uranium-238 has a half-life of 4.468 billion years. Are We There Yet? Did you know that a light-minute is a measure of distance, not time? In August 2012 and November 2018, Voyager 1 and 2 crossed the heliopause, the boundary where the Sun's solar wind meets the interstellar medium. Using NASA's Deep Space Network , Earth communicates with both spacecraft and tracks each on its journey. At the time of this article, the Canberra antenna tracked Voyager 1 at 25.80 billion km, or just

November's Night Sky Notes: Light-Day Away

over 16 billion miles. This graphic shows the position of NASA's Voyager 1 and Voyager 2 probes, outside of the heliosphere, a protective bubble created by the Sun that extends well past the orbit of Pluto. NASA/JPL-Caltech On November 18, 2026, 49 years after its launch, Voyager 1 will cross the one-light-day mark, 16.1 billion miles from the Sun. You may have heard the term light-year in your favorite sci-fi movie or show, or when stargazing and referring to an object outside of our solar system. This video best demonstrates the different distances between the inner solar system, outer solar system, and beyond, using light measurements. Additionally, track each Voyager spacecraft here . You can get a real understanding of what a gravity assist might be like with our Gravity Assist activity and learn what it's like to measure a "light minute" with our Measuring a Kid Minute activity guide!

THE CHRONOLOGY

Signals across the window

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A NOTE ON THIS EDITION

The reporting window

7 stories / the previous 48 hours

This issue brings together reports from the Publisher's approved source records. Each story remains attributed to its originating source. The editorial layout organises the material for a calm, uninterrupted reading experience; it does not replace the original reporting or add unverified claims. Image credits and usage notes are retained with the relevant material and in the printed source pages.

The universe is not short of stories. The task is to give each one the room it deserves.

PROVENANCE

Sources & image credits

01 / NASA News

APOD: 2026 October 31 - Night on a Spooky Planet

https://science.nasa.gov/image-article/apod-2026-october-31-night-on-a-spooky-planet/

IMAGE CREDIT: Stéphane Vetter (Nuits sacrées)

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02 / NASA News

NASA Releases MAX POWER Key Events for Media, Participants

https://www.nasa.gov/news-release/nasa-releases-max-power-key-events-for-media-participants/

IMAGE CREDIT: NASA

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03 / NASA Space Station

Space Crops and Advanced Tech Wrap Up Week Following Spacewalk

https://www.nasa.gov/blogs/spacestation/2026/10/30/space-crops-and-advanced-tech-wrap-up-week-following-spacewalk/

IMAGE CREDIT: NASA

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04 / NASA Image of the Day

NASA's James Webb Space Telescope Captures Eerie Cat's Head

https://www.nasa.gov/image-detail/full-res-for-display-6/

IMAGE CREDIT: NASA

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05 / NASA News

NASA's James Webb Space Telescope Captures Eerie Cat's Head

https://www.nasa.gov/image-article/nasas-james-webb-space-telescope-captures-eerie-cats-head/

IMAGE CREDIT: NASA

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06 / NASA News

NASA Explores Potential Commercial Servicing for Hubble Telescope

https://science.nasa.gov/missions/hubble/nasa-explores-potential-commercial-servicing-for-hubble-telescope/

IMAGE CREDIT: Not supplied

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07 / NASA News

November's Night Sky Notes: Light-Day Away

https://science.nasa.gov/solar-system/skywatching/night-sky-network/novembers-night-sky-notes-light-day-away/

IMAGE CREDIT: Not supplied

KEEP LOOKING UP

The next question is waiting.

A source-linked window on the universe. Follow the reporting, question the evidence and keep exploring.